

Bakhrom Reyimberganov

Full-Stack & IoT Engineer

Engineer who builds the whole path from sensor to dashboard — ESP32 firmware in C++, async FastAPI services on PostgreSQL, and React, Electron or Android clients on top. Most of my work is production systems for utilities, agriculture and campus infrastructure.

Experience

2024 — now

Uzbekistan /
remote

Independent Software Engineer

Self-employed — production systems for industrial and health clients

- **Ugaz ARM** — operator workstation and central server for CNG filling stations, in C#/.NET over EF Core and PostgreSQL. Three dispenser drivers behind one interface, including a simulator that makes the system testable without hardware.
- **TE71 Meter Monitor** (Toshelectroapparat) — one async FastAPI backend covering electricity, water and gas. ESP32 meters post behind per-device tokens; firmware ships as OTA batch rollouts.
- **EX518 DLMS tester** — Python tool speaking DLMS/COSEM over USB-RS485 to Holley three-phase meters, reading registers and driving relay disconnect.
- **SADO** — speech-therapy platform on async FastAPI, Celery and MinIO, running Whisper transcription into an XGBoost risk classifier.
- **ECO COMPANY** — FastAPI site backend with JWT auth, a DeepSeek chatbot over FastMCP and Telegram admin alerts.

C# · .NET · Python · FastAPI · PostgreSQL · ESP32 · Celery · Docker

Selected projects

2026

Uni-Nav — indoor campus navigation

Wayfinding system for university buildings, built on one API with four clients: web, Electron kiosk and an offline Android app. Buildings are modelled as a graph and routed with A*, including transitions between floors.

Python · FastAPI · PostgreSQL · TypeScript · React · Electron · Kotlin

2026

WMAX — remote patient monitoring

Early-warning system for discharged patients, built for the National AI Hackathon in Khorezm. A smartwatch streams vitals to the backend, which compares them against each patient's own circadian baseline rather than a fixed threshold.

Python · FastAPI · PostgreSQL · React · Kotlin · Flutter · Wear OS

2026

AI Art Evaluation System

Web app that scores artwork across five weighted categories using Gemini and returns per-category feedback. Live at analyst-omega-red.vercel.app.

Python · Django REST · PostgreSQL · TypeScript · React · Gemini API

2026

Smart Greenhouse

Climate and irrigation controller that drives pumps, fan and lighting from per-sensor thresholds. The MQTT topic contract was published first, so the backend and dashboard were built against a fixed interface.

C++ · ESP32 · MQTT · Python · FastAPI · TypeScript

2026	Network traffic simulator SDN testbed that generates labelled network datasets for machine learning — four topologies, eleven routing schemes, and eight attack patterns on a daily load curve. Python · Mininet · Open vSwitch · SDN · Docker
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Technical skills

Languages	Python, TypeScript, C / C++, Kotlin, Dart, SQL
Backend	FastAPI, Django REST, async SQLAlchemy, PostgreSQL, WebSocket, MQTT, JWT auth, Docker Compose, CI
Frontend	React, Vite, Tailwind, shadcn/ui, react-query, Recharts, Fabric.js, PWA
Mobile	Flutter, Jetpack Compose, Wear OS, Electron, SQLite
Embedded	ESP32, Arduino-ESP32, PlatformIO, RS-485, ADC sensors, TFT_eSPI, OTA updates
Also	Gemini API, LLM agent orchestration, A* pathfinding, Mininet / SDN, Asterisk PBX, Sentry
Spoken	Uzbek (native), English (B1), Russian (B1)

Education

2023 — 2027 expected	BSc, Software Engineering Urgench State University — transferred in 2025 from Tashkent University of Information Technologies when its regional branch closed Academic projects: campus IP telephony on Asterisk PBX (extensions, IVR, ring groups, Docker Compose) · clinical decision support for bronchial asthma (rule engine + baseline ML with a structured explanation layer)
2021 — 2023	Academic Lyceum Urgench State University

Competitions

2026	ICPC — semifinalist International Collegiate Programming Contest
2026	13th of 60 teams — AI Healthcare Hackathon, CAU Team entry — medical image classification and segmentation in PyTorch, served through Streamlit
2026	Best Agro Project — A500 Hackathon <u>AgriGuard AI</u> — detects crop pests and targets them with directed UV light instead of chemicals
2023	Two bronze medals — OMOUS Open Mathematical Olympiad among University Students, Turkmenistan